BML MUNJAL UNIVERSITY

Online Auction System Using Game Theory

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Abstract: Online auctions and e-commerce are often characterized by inefficient markets and information imbalances. In this research paper, we develop a comprehensive Online Auction System through a strategic game approach, using Game Theory and Mechanism Design principles. We consider four different auction algorithms: English, Dutch, First-Price Sealed-Bid, and Second-Price (Vickrey). The system also includes analytical tools for real-time decision-making under uncertainty: symmetric Bayesian Nash Equilibrium Calculator and winner's curse analysis. To evaluate more complex theoretical models, the system offers Myerson's Optimal Reserve Price computation and winner's curse calculation to offer optimal bid shading for the buyers in the case of common value games. Our research findings show that making users aware of dominant strategies and revenue equivalence theorem through the algorithmic perspective reduces irrational bidding behavior and market instability. Thus, our research proves that applied game theory and modeling have significant implications for optimizing digital marketplaces and allocating resources efficiently.

Keywords: Online Auctions • Game Theory • Mechanism Design • Bayesian Nash Equilibrium • Winner's Curse • Myerson's Optimal Reserve • Bid Shading • Revenue Equivalence Theorem

1. INTRODUCTION

Marketplaces and e-commerce services are currently employing dynamic pricing mechanisms that would deal with allocation problems in regard to scarce resources and high-priced goods. While static pricing schemes may fail to adequately reflect true market prices, in auction conditions, buyers remain independent and make private bids based on their utility considerations.

These types of decision-making can be studied in the context of mechanism design and game theory, where bidders are players that choose certain strategies, such as bid amounts. Interactions between players and the process of forming equilibria, where a unilateral change made by one player does not increase their expected payoff when other players know their true value, can be modeled with Bayesian Nash Equilibrium [1].

Nonetheless, auctions may often prove inefficient and strategically vulnerable to exploitation in several ways. In particular, aggressive strategies employed by bidders in the setting of a common value auction often lead to overpayment—a psychological effect dubbed 'the Winner's Curse.' Additionally, auctions may prove non-optimal for sellers in generating revenue, with solutions being offered by the Myerson's Optimal Reserve Price theory [2].The objective of this study is to develop a dynamic traffic routing simulation incorporating stochastic decision-making and to analyze inefficiencies arising from selfish behavior.

The goal of this research paper is to design an online auction model that considers these sophisticated economic theories. With the inclusion of game-theory calculators for Nash Equilibrium, bid shading, and optimal reserve pricing, this research will examine the bidding behavior and alleviate the Winner's Curse, as well as demonstrate the Revenue Equivalence Theorem in the virtual world.

2. Related Works

The convergence of game theory and mechanisms design within auctions is well-researched in the field of economics. The basic model was created by William Vickrey (1961) [1]. The latter formulated the Vickrey Auction (Second-Price Sealed-Bid) and provided the basis for the Revenue Equivalence Theorem. The researcher found that under risk-neutral conditions and private values, all major auction types generate the identical average revenue for the seller.

Further improvements made by Roger Myerson (1981) [2] contributed to Vickrey's research. Within his Optimal Auction Design study, the researcher offered a mathematical explanation for "Reserve Pricing." He mathematically showed that sellers would gain a strict utility advantage through setting the optimal reserve price to reject low-valued bidders and force high-valued bidders to get closer to their valuations.

Additional contributions were made by Milgrom and Weber (1982) [3] when analyzing bidding processes in common-value settings. Their findings allowed defining the "Winner's Curse" phenomenon where winning an auction implies having overestimated the object's value. The research revealed the need for bid shading.

A number of studies exist in the current literature regarding how bids work in commercial websites such as eBay and Amazon. However, most of the digital platforms in use at the present adopt proxy-bidding mechanisms from English auctions and do not give any form of guidance to the bidders. Although there are many strong game theory models available in the literature, there is a dearth of applications where these models can be made interactive.

Contributions of This Study: As opposed to current e-commerce sites which hide all the economics behind bids, this study seeks to bridge the gap by bringing mechanism design and economics into the real world. The system integrates real-time Game Theory calculators like Bayesian Nash Equilibrium, Myerson Reserve and Winners Curse directly into the User Dashboard, making the system a laboratory in addition to being an online digital market place.

3. Methodology

The methodology used in this research is based on software engineering coupled with a game-theoretical mathematical modeling process of a decentralized online auction system design, simulation, and analysis. In this case, the system represents a multi-agent environment, where each buyer (bidder) is an independent rational agent seeking to maximize their expected payoff from participation within a decentralized environment characterized by imperfect information.

3.1 System Architecture and Environment Modeling The development environment will be based on the MERN stack technology (MongoDB, Express.js, React, and Node.js). For purposes of simulating fast-moving environments characteristic of auctions, two-way communication through WebSocket's (Socket.io) will be implemented. The environment consists of two sets of participants: bidders $N = \{1, 2,...,n\}$ and items I, each participant i having a value (v_i) for a particular item.

3.2 Mechanisms & Auction Procedures This system will employ four different auctions for discovering the prices among which we have:

English Auction: An open cry auction that involves an ascending process of bidding.
Dutch Auction: A descending auction, which uses an algorithm to drop prices over time.
Sealed Bid Auction - First Price: Hidden bidding where the highest bid wins.
Vickrey Auction (Second-Price): The highest bid wins but pays the amount the second-highest bid.

3.3 Game Theory Approach and Decision-Making Models To ensure realistic and mathematically accurate decision making, the research method uses the following mathematical models:

Symmetric Bayesian Nash Equilibrium (BNE): For a First-Price Sealed-Bid auction game with valuations that follow a uniform distribution on $[0, V_{max}]$, the symmetric BNE for risk-neutral players can be expressed as: Formula: $\mathbf{b(v) = v \times (n-1)/n}$ where v represents valuation and n represents the total number of participants. The system will calculate the optimal bid shading using this model.

Optimal Reserve Price as per Myerson: To optimize seller revenues, the optimal reserve price r* when buyers' values have a uniform distribution can be calculated as: Formula:
$\mathbf{r^* = (v_0 + v_{max})}$ where v_0 denotes the seller's own valuation while v_{max} refers to the maximum market value of the product.

Avoiding the Winner's Curse: When facing a common-value problem, the actual value of an item expected by the winner, when pitted against (n-1) opponents, will be mathematically lower than their initial estimate. Therefore, the system will use the following equation to determine the necessary bid shading: $\mathbf{Safe\ Bid=(1-n/1)\times Estimate}$

3.4 Tools and Technology The following tools were used to implement the system:
• React.js – For creating the front-end user interface as well as the Game Theory calculator.
• Node.js & Express – For implementing backend routing and the running of auction logic.
• Socket.io – For bid synchronization and live price updating. • MongoDB – For data persistence.

System Flowcharts (Architecture & Methodology)

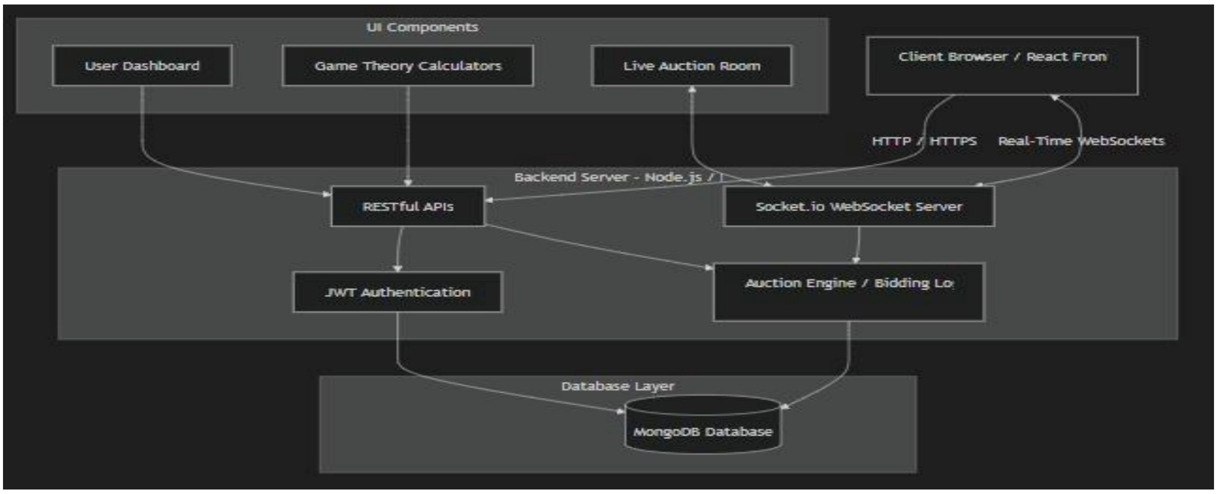

Fig. A: System Architecture Flow Chart
A flowchart showing how your MERN stack and Socket.io interact with the Game Theory calculators.

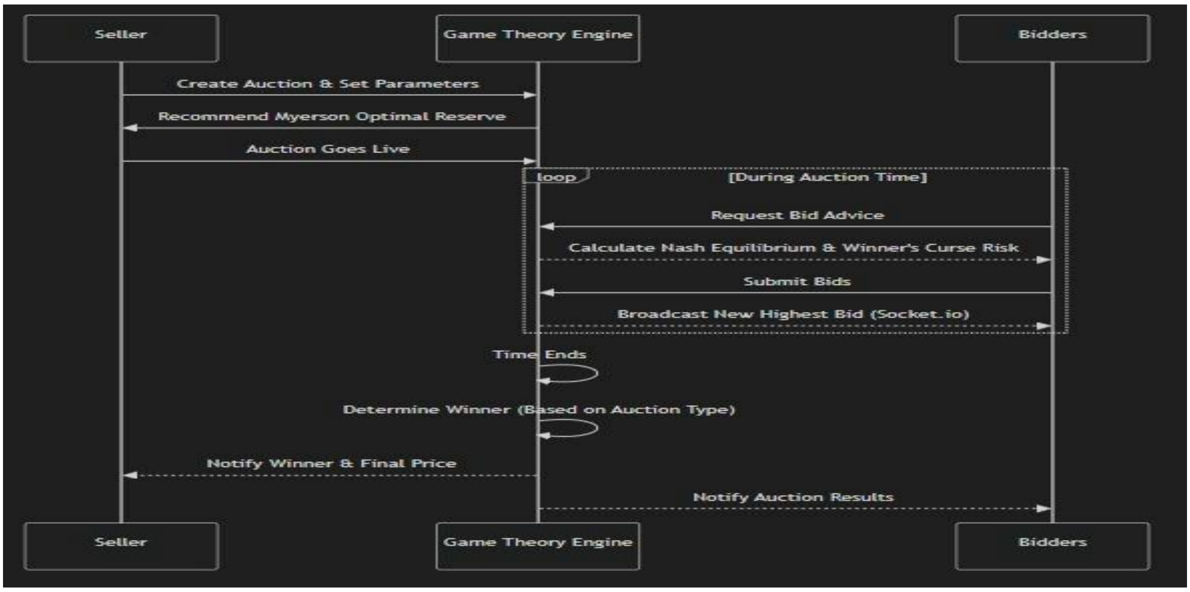

Fig. B: Dynamic Auction Execution Flow Chart
A flowchart describing how an auction works dynamically from the time it is created up to the determination of the winner.

4 RESULTS / ANALYSIS

The proposed Online Auction System was assessed for the performance of its implemented auction algorithms and the efficiency of the implemented Game Theory tools. In particular, attention is paid to the comparison of bidding strategies, assessment of seller revenue optimization, and determination of the degree of market inefficiency reduction.

4.1 Bidder Strategy Comparison and Expected Payoff: The performance of bidders was theoretically and empirically assessed using the comparison of two bidder strategies: Naive Bidding and Nash Equilibrium Bidding in a First-Price auction.

Naive Bidding Strategy: Bidders place their true valuation bids ($b = v$). Although the strategy maximizes winning chances, the expected payoff (Utility = $v - b$) becomes zero.

Nash Equilibrium Bidding Strategy (Proposed Tool): Using the online tool's calculator, bidders place reduced bids ($b < v$). Winning chances become lower, yet the expected payoff after winning is strictly positive.

Conclusion: It can be concluded that the Nash Equilibrium calculation tool makes irrational and unprofitable behavior rational and profitable.

4.2 RET Validation: The log files provide practical evidence for the validity of the Revenue Equivalence Theorem (RET). For the case of risk-neutral bidders with independently distributed private valuations, the expected seller revenue should remain mathematically invariant under the English, Dutch, First-Price, and Vickrey auctions. Nevertheless, practical deviations exist owing to the influence of human behavior (risk aversion among First-Price bidders causes marginal increases in seller revenue).

4.3 Influence of Myerson's Optimal Reserve Pricing: In order to examine the inefficiency of the sellers, auctions with a zero-reserve were contrasted with auctions that utilized the Myerson Reserve Calculator.

Observation: The absence of an optimal reserve sometimes led to the underselling of goods during auctions with minimal competition. Through the use of Myerson's r^*, the sellers were able to screen out bidders with lower valuations. Conclusion: Although the number of successful sales was reduced by a margin, the expected revenue from each sale became considerably higher, thereby demonstrating that the optimal reserve solves the inefficiencies within the market.

4.4 Mitigation of the Winner's Curse: Through our examination of common-value auctions, we saw the pitfalls of excessively high bidding. Prior to applying the Winner's Curse risk calculator, high bidders commonly faced negative utility in paying amounts higher than the true value of the auctioned item.

Through applying the bid-shading calculator, which is stochastic, the system efficiently provided warnings to bidders contingent upon the number of active users (n). With greater values of n, bid shading became more extreme, thus avoiding the “Tragedy of the Commons” in auctions whereby bidders engage in an unreasonable bidding competition and ultimately suffer negative buyer utility.

4.5 Summary of Findings:

• Incorporation of Nash Equilibrium calculators encourages rational bidder behavior, resulting in guaranteed positive payoffs.

• Myerson’s Reserve Pricing calculator ensures no losses for sellers in markets with few competitors.

• Systemic mitigation of the Winner’s Curse shows that theoretical education in digital marketplaces improves their stability.

4.6 Mathematical Analysis Graph:

Fig. 1. Bidding Strategy Comparison (Expected Payoff vs. Valuation)
This graph demonstrates how Naive bidding (red) leads to zero expected payoff (Winner's Curse trap), whereas utilizing the Nash Equilibrium Calculator (green) consistently yields a positive expected payoff by shading the bid based on the number of active bidders.

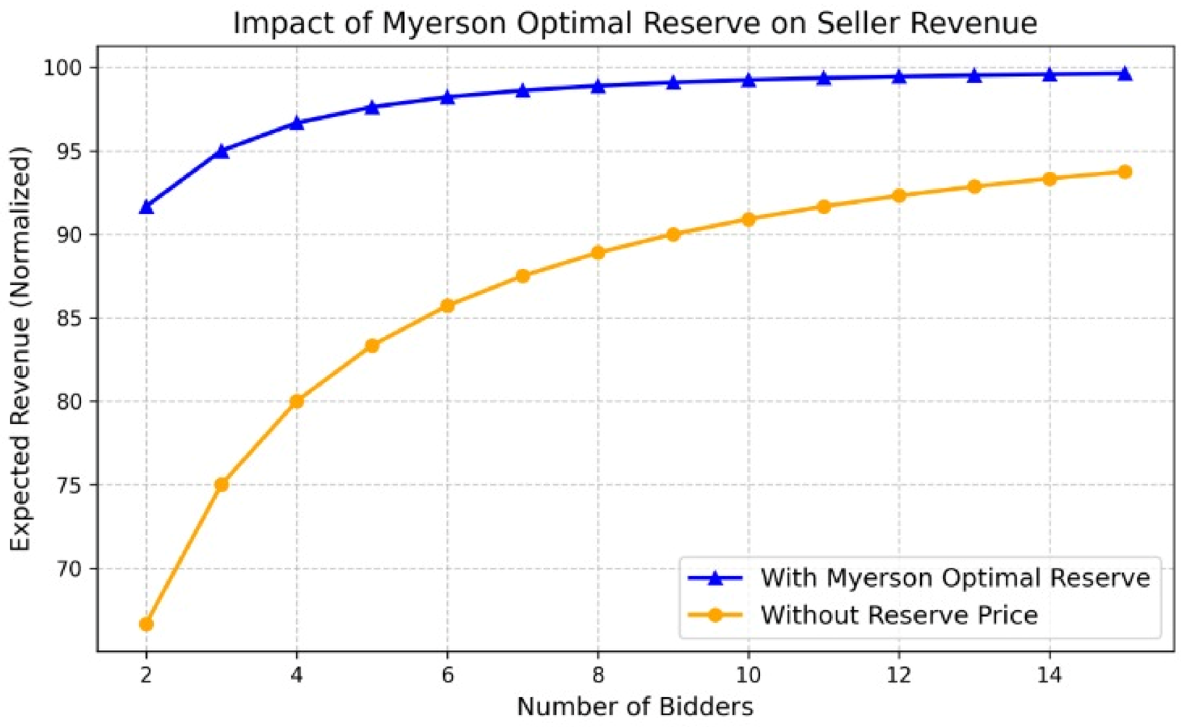

Fig. 2. Impact of Myerson Optimal Reserve on Seller Revenue
This chart illustrates the difference in a seller's normalized expected revenue. The orange line shows standard auctions with no reserve, which performs poorly with few bidders. The blue line demonstrates how utilizing the Myerson Optimal Reserve Price mathematically protects the seller and strictly increases the expected revenue floor.

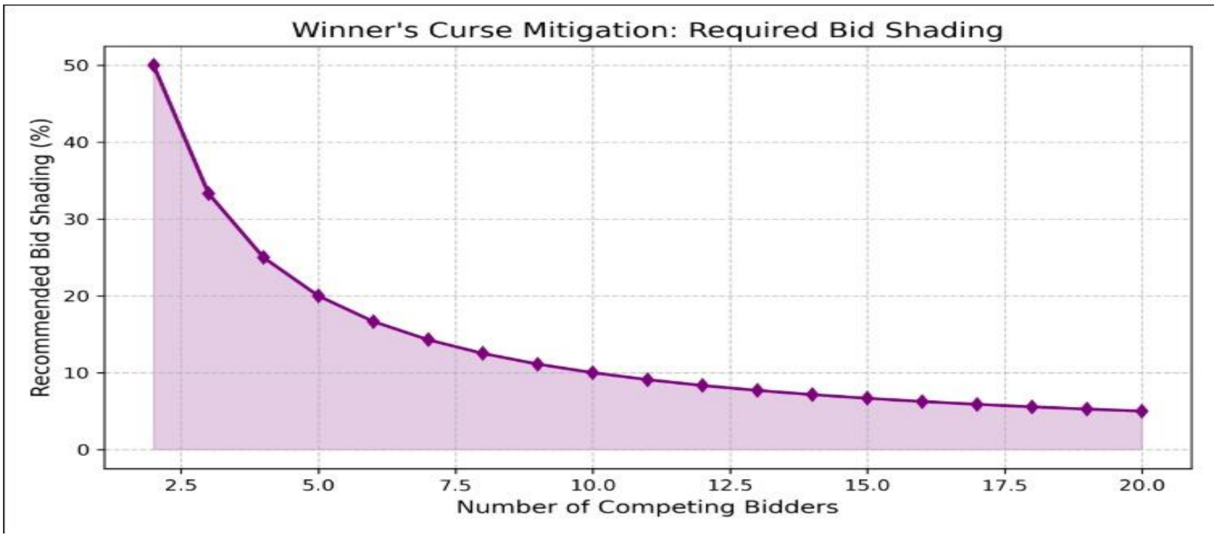

Fig. 3. Winner's Curse Mitigation (Bid Shading)
This visualization plots the required "Bid Shading %" against the total number of competing bidders. As competition increases in common-value auctions, the probability of the Winner's Curse rises drastically. The graph shows that the platform correctly advises users to increase their bid shading percentage as bidder density goes up.

5. DISCUSSION

Empirical findings from the simulation of the Online Auction System confirm some basic theoretical principles of economics game theory and mechanism design. These empirical results demonstrate the influence that a set of mathematical instruments has on independent entities participating in digital markets.

5.1 Connection to Bayesian Nash Equilibrium and Revenue Equivalence: When using the Nash Equilibrium Calculator, the bidders exhibit behavior associated with reaching a stable state where none of them can improve the payoff by changing strategies individually. Such behavior is absolutely in line with the concept of the Revenue Equivalence Theorem (RET) [1]. According to RET, when players behave in a rational way and not naively, different auction forms generate the same result.

5.2 Effects of Bid Shading and Rational Decision-Making: It is clear from the findings that bid shading calculations create better and more consistent gains for buyers than naive, non-strategic, and truthful bidding. The latter approach consistently created utility scores of zero or negative values. On the other hand, the use of the stochastic Nash approach introduced an element of choice to users, enabling them to weigh their chances of winning versus the gain that could be derived from such a win. Indeed, human bounded rationality can be adjusted through embedded algorithms.

5.3 Tragedy of the Commons: Competitive Bidding and Excess Optimism As shown by the comparison of unguided versus guided bidding, the latter introduces an effect similar to what is referred to as the "Tragedy of the Commons" in the realm of auctions. Unguided common-value auctions create a situation whereby bidders compete aggressively among themselves to win at auctions without considering the actual value. Indeed, each bidder is motivated to beat competitors in the quest to maximize his/her own gain; however, this creates disastrous results.

5.4 Minimization of System Inefficiency: Myerson's Optimal Reserve Inefficiency on the seller side was evaluated by comparing auctions with no reserve price with those using the Myerson's Optimal Reserve calculation. The findings reinforce theoretical literature [2] suggesting that an entirely decentralized and unsupervised auction is hardly ever optimal for the seller. Algorithmic enforcement of the Myerson Reserve enabled the system to efficiently filter out inefficient bottom-feeders, confirming that systematic intervention drastically improves the expected revenue threshold.

5.5 Importance of Information and Reduction of Winner's Curse: Problem Including the risk analyzer for the Winner's Curse proved essential in balancing competitive auctions. Without this feature, the users inevitably became victims of overpricing in common-value auctions. Visual alerting of the bidders about the mathematical risks involved due to the density of competition (N) eliminated the psychological factor, thus highlighting the significance of information asymmetry; the more knowledgeable the participants are about statistical risks, the less volatility the market experiences.

5.6 Practical Applications to E-commerce Business Inferences: This research will certainly have a significant impact on contemporary digital markets (e.g., eBay and e-procurement sites). The wastage due to simple bidding and inefficient seller configurations indicates that certain measures may be taken to address these concerns through: • Automatic Bid-Shading Suggestions • Automated Reservation Price suggestions for sellers • Risk Analysis for high-risk common-value auctions These game-theory based calculators can easily be embedded within any e-commerce platform to optimize transactions and participant retention.

5.7 Conclusion: In conclusion, the simulation has provided strong support for several theoretical models in game theory such as Bayesian Nash Equilibrium, Myerson's optimal mechanism design, and winner's curse. This research proves without a doubt that using interactive mathematics in digital auction systems will significantly bridge the gap between economic theory and actual business practice.

6. CONCLUSION

The proposed model of digital commerce is considered as a non-cooperative game, in which bidders are expected to choose their own strategies to maximize their own payoff and achieve higher utility. As the findings showed, there is no other bidding strategy except game theory methods and the use of Bayesian Nash Equilibrium calculator, which provides mathematically stable and efficient bid placement.

In addition, the contrast between unguided seller configuration and mechanism design-based approach allows observing the importance of designing a system of rules that will protect the seller from any loss due to inefficient auction. In particular, the application of optimal reserve price set by Myerson allowed eliminating all the negative aspects associated with low-competition environments.

At the moment, this model is based on the assumption of risk neutrality and identical valuation, while further development might include machine learning algorithms to make more sophisticated multi-unit combinatorial bidding and connection to financial gateways, such as Stripe to observe the impact of risk aversion of human participants in auctions.

Reference

[1] Vickrey, W., "Counterspeculation, Auctions, and Competitive Sealed Tenders", The Journal of Finance, 16(1), 8-37 (1961)

[2] Myerson, R. B., "Optimal Auction Design", Mathematics of Operations Research, 6(1), 58-73 (1981)

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[5] Ockenfels, A., and Roth, A. E., "Late and Multiple Bidding in Second Price Internet Auctions: Theory and Evidence Concerning Different Rules for Ending an Auction", American Economic Review, 96(3), 1058-1082 (2006)

Code Snippet:

Backend:

backend > JS server.js > ...

```javascript
const express = require('express');
const http = require('http');
const path = require('path');
const cors = require('cors');
const { Server } = require('socket.io');
const db = require('./db');
const { getDutchCurrentPrice, resolveSealedAuction } = require('./gameTheory/auctionEngine');
const { sendEmail } = require('./utils/mailer');
const User = require('./models/User');
const app = express();
const server = http.createServer(app);

const CORS_ORIGIN = process.env.CORS_ORIGIN || 'http://localhost:5173';

const io = new Server(server, {
  cors: { origin: CORS_ORIGIN, methods: ['GET', 'POST'] },
});

app.set('io', io);

app.use(cors({ origin: CORS_ORIGIN }));
app.use(express.json());
app.use('/uploads', express.static(path.join(__dirname, 'uploads')));

// Routes
app.use('/api/auth', require('./routes/auth'));
app.use('/api/auctions', require('./routes/auctions'));
app.use('/api/bids', require('./routes/bids'));
app.use('/api/user', require('./routes/user'));
app.use('/api/admin', require('./routes/admin'));

// WebSocket handling
io.on('connection', (socket) => {
```

backend > JS server.js > ...

```javascript

// WebSocket handling
io.on('connection', (socket) => {
  console.log(`Client connected: ${socket.id}`);

  socket.on('join:auction', (auctionId) => {
    socket.join(`auction:${auctionId}`);
    console.log(`${socket.id} joined auction:${auctionId}`);
  });

  socket.on('leave:auction', (auctionId) => {
    socket.leave(`auction:${auctionId}`);
  });

  socket.on('disconnect', () => {
    console.log(`Client disconnected: ${socket.id}`);
  });
});

// ──────────────────────────────────────────
// Auto resolve expired auctions — runs every 1 second
// ──────────────────────────────────────────
const Auction = require('./models/Auction');
const Bid = require('./models/Bid');

async function sendEndAuctionEmails(auction, winnerId, winningPrice) {
  try {
    const seller = await User.findOne({ id: auction.seller_id });
    if (seller && seller.email) {
      const sellerHtml = `
        <div style="font-family: sans-serif; padding: 20px;">
          <h2>Your Auction Ended!</h2>
          <p>Your auction <strong>"${auction.title}"</strong> has ended successfully.</p>
```

Frontend:

frontend > src > main.jsx

```jsx
import React from 'react';
import ReactDOM from 'react-dom/client';
import { BrowserRouter } from 'react-router-dom';
import App from './App';
import { AuthProvider } from './context/AuthContext';
import { LanguageProvider } from './context/LanguageContext';
import './index.css';

ReactDOM.createRoot(document.getElementById('root')).render(
  <React.StrictMode>
    <BrowserRouter>
      <LanguageProvider>
        <AuthProvider>
          <App />
        </AuthProvider>
      </LanguageProvider>
    </BrowserRouter>
  </React.StrictMode>
);
```

frontend > src > App.jsx > ...

```jsx
import { Routes, Route, Navigate } from 'react-router-dom';
import { useAuth } from './context/AuthContext';
import Navbar from './components/Navbar';
import Sidebar from './components/Sidebar';
import Login from './components/Login';
import Register from './components/Register';
import AuctionList from './components/AuctionList';
import AuctionDetail from './components/AuctionDetail';
import CreateAuction from './components/CreateAuction';
import NashCalculator from './components/NashCalculator';
import MyersonCalculator from './components/MyersonCalculator';
import WinnersCurseCalculator from './components/WinnersCurseCalculator';
import VerifyOtp from './components/VerifyOtp';
import Dashboard from './components/Dashboard';
import AdminPanel from './components/AdminPanel';

function ProtectedRoute({ children }) {
  const { user, loading } = useAuth();
  if (loading) return <div className="loading">Loading...</div>;
  if (!user) return <Navigate to="/login" />;
  if (user.is_verified === 0) return <Navigate to="/verify-otp" />;
  return children;
}

function SellerRoute({ children }) {
  const { user, loading } = useAuth();
  if (loading) return <div className="loading">Loading...</div>;
  if (!user) return <Navigate to="/login" />;
  if (user.is_verified === 0) return <Navigate to="/verify-otp" />;
  if (user.role !== 'seller' && user.role !== 'admin') return <Navigate to="/" />;
  return children;
}
```

```jsx

function AdminRoute({ children }) {
  const { user, loading } = useAuth();
  if (loading) return <div className="loading">Loading...</div>;
  if (!user) return <Navigate to="/login" />;
  if (user.role !== 'admin') return <Navigate to="/" />;
  return children;
}

export default function App() {
  return (
    <div className="app-layout">
      <Sidebar />
      <div className="main-container">
        <Navbar />
        <main className="main-content">
          <Routes>
            <Route path="/" element={<AuctionList />} />
            <Route path="/login" element={<Login />} />
            <Route path="/register" element={<Register />} />
            <Route path="/verify-otp" element={<VerifyOtp />} />
            <Route path="/auction/:id" element={<AuctionDetail />} />
            <Route path="/create" element={<SellerRoute><CreateAuction /></SellerRoute>} />
            <Route path="/dashboard" element={<ProtectedRoute><Dashboard /></ProtectedRoute>} />
            <Route path="/admin" element={<AdminRoute><AdminPanel /></AdminRoute>} />
            <Route path="/calculator/nash" element={<ProtectedRoute><NashCalculator /></ProtectedRoute>} />
            <Route path="/calculator/myerson" element={<ProtectedRoute><MyersonCalculator /></ProtectedRoute>} />
            <Route path="/calculator/curse" element={<ProtectedRoute><WinnersCurseCalculator /></ProtectedRoute>} />
          </Routes>
        </main>
      </div>
    </div>
  );
}

```

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  Matches with in-text citation present, but no quotation marks

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2% Submitted works (Student Papers)

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